표본평균과 표본분산의 도출

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1. 모형의 설정 $\hat{\mu} = E(y_i) = \frac{\sum_{i=1}^{n} y_i}{n}, \hat{\sigma}^2 = \frac{\sum_{i=1}^{n} y_i}{n}$

- 모평균 → 확률변수 yi가 취할 수 있는 모든 값의 평균이
  μ : $E(y_i) = \mu$, 모분산 → 분산이 σ^2 : $E((y_i - \mu)^2) = \sigma^2$ 이라고 할 때, 다음을 모형으로
  → Population Equation : $y_i = \mu + e_i (=$ 잔차항$)$ (i=(1,2, ... ,N); 고전적 가정) with,
  - 1) $\forall i, E(e_i) = 0. (\because y_i - \mu = e_i, E(y_i - \mu) = E(e_i) = 0;$
    2) $E(e_i^2) = \sigma^2, \forall i$(var def 대입); 3) $E(e_i e_j) = 0, \forall i, j$ (cov def 대입)

2. 최소제곱법(OLS) : 잔차항의 제곱의 합을 극소화 하는 방법 (μ추정 방식 중 하나)

- 표본회귀식 : $y = \hat{\mu} + \hat{e}_i$ ($\hat{\mu} \neq \mu$ in real life)
- OLS추정량 유도 → $min \sum_{i=1}^{n} \hat{e}_i^2$ (with respect to $\hat{\mu}$) (F.O.C.) $\frac{d \sum \widehat{e_i}^2}{d\hat{\mu}} = 0 \rightarrow \hat{\mu} = \frac{1}{n} \sum Y_i$

3. OLS 추정량의 불변성(unbiasedness) : $E(\hat{\theta}) = \theta$ (성립시 unbiased estimator)

- $E(\hat{\mu}) = (n^*\mu$ 방식$) = ($pop e.q. 방식$) = \mu$

Repeated sampling 에서 평균적으로 θ의 값을 준다

4. OLS 추정량의 분산+ 가정 : $E(e_i e_j) = 0, i \neq j$

- $Var(\hat{\mu}) = E\left[\left(\hat{\mu} - E(\hat{\mu})(= \mu)\right)^2\right] = E\left[\left(\frac{\sum e_i}{n}\right)^2\right] = \frac{\sigma^2}{n} \quad \left(E(e_i^2) = \sigma^2, \forall i\right)$

5. 표본 분산의 평균

- 1) known μ : $\hat{\sigma} = \frac{1}{n} \sum_{i=1}^{n} e_i^2$
  $recall\, E(e_i^2) = \sigma^2$ & w.t.s. $E(\widehat{\sigma^2}) = \sigma^2 (u.b.e.), E\left(\frac{\sum e_i^2}{n}\right) = \sigma^2$
  $(pop\ e.q.)\, Y_i = \mu + e_i \rightarrow \widehat{\sigma^2} = \frac{1}{n} e_i^2 = \frac{1}{n} \sum (Y_i - \mu)^2$
- 2) unknown μ : $\widehat{\sigma^2} = \frac{1}{n-1} \sum_{i=1}^{n} e_i^2$

$$e_i = (Y_i - \hat{\mu}) + (\hat{\mu} - \mu) = \hat{e}_i + (\hat{\mu} - \mu)$$
$$E\left(\sum_{i=1}^{n} e_i^2\right) = E\left(\sum_{i=1}^{n} \left(\hat{e}_i + (\hat{\mu} - \mu)\right)^2\right) = E\left(\sum \widehat{e_i}^2\right) + \sigma^2$$

6. 표본분산의 분산에 대한 추정 :$Y_i = \mu + e_i, e_i \sim iid\, N(0, \sigma^2), E(\bar{y}) = \mu, Var(\bar{y}) = \frac{\sigma^2}{n}$

- y의 분포? 1) σ^2 is known, 2) σ^2 is unknown

→ 1) σ is known : $let\ Z_i = \frac{e_i}{\sigma} \sim N(0,1); \quad Z_1^2(\sim \chi^2) + \cdots + Z_n^2 \rightarrow \sum_{i=1}^{n} \left(\frac{e_i}{\sigma_i}\right)^2 \sim \chi^2(n)$

$$= \sum_{i=1}^{n} Z_i^2 \sim \chi^2(n); \quad e_i = y_i - \mu \rightarrow \hat{e}_i = y_i - \hat{\mu} \rightarrow \frac{\sum \hat{e}_i^2}{\sigma^2} \sim \chi^2(n-1)$$

→ 2)σ is unknown: $\hat{e}_i = y_i - \hat{\mu}, \frac{\sum \hat{e}_i^2}{\sigma^2} \sim \chi^2(n-1); \quad \widehat{e_i^2} \rightarrow \frac{\sum \hat{e}_i^2}{n-1} = \widehat{\sigma^2}, \frac{(n-1)\widehat{\sigma^2}}{\sigma^2} \sim \chi^2(n-1)$

$E(\widehat{\sigma^2}) = \sigma^2, \sigma^2 = ? \quad {**}\ Note\!:\ X \sim \chi^2(p), E(X) = p, Var(X) = 2p$

$$E\left(\frac{(n-1)\widehat{\sigma^2}}{\sigma^2}\right) = n - 1, E(\hat{\sigma}) = \widehat{\sigma^2}; \quad Var\left(\frac{(n-1)\widehat{\sigma^2}}{\sigma^2}\right) = 2(n-1)$$
$$\rightarrow \frac{(n-1)^2}{\sigma^4} Var(\widehat{\sigma^2}) = 2(n-1), var(\widehat{\sigma^2}) = \frac{2\sigma^4}{n-1}$$

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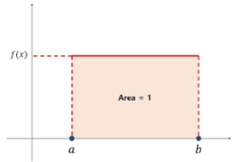
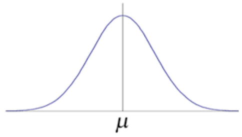
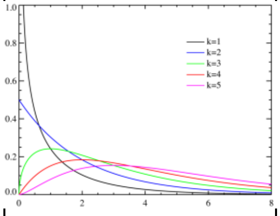
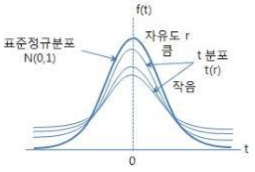
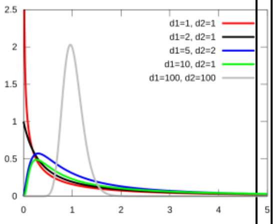
확률분포이론











1. 균일 분포 : r.v. X→ [a,b]에서 일정 확률을 가지고 있을 때 X~U(a,b)

- $CDF:\ \frac{1}{b-a}\ (a \le x \le b)$; 0 (otherwise); E(X) = $\int_a^b \frac{1}{b-a} x\ dx = \frac{a+b}{2}$; Var(X) = $\frac{(b-a)^2}{12}$

$E(X) = \int_a^b f(x)x\ dx,$

$Var(X) = \int_a^b (X - E(X))^2 f(x)\, dx$

2. 정규분포 : r.v. X → 정규분포를 따른다 : X~N(μ,σ²)

- $CDF: f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$; X~N(μ,σ²)→ aX+b~N(aμ+b,a²σ²)

  → 표준정규분포 : X~N(μ,σ²) → Z=(X-μ)/σ ~ N(0,1); $CDF: f(z) = \frac{1}{\sqrt{2\pi}} exp\left(-\frac{(z)^2}{2}\right)$

  $E(X) = \int_{-\infty}^{\infty} xf(x)\ dx = 0, \quad Var(X) = \int_{-\infty}^{\infty} \left(X - f(X)\right)^2 f(x)\, dx = 1$

3. χ² 분포(카이제곱분포) : Z~N(0,1)→Z²~χ²(1) (1→자유도)

- $X_1, X_2, \cdots, X_P \sim iid\ N(0,1) \rightarrow X_1^2 + \cdots X_p^2 \sim \chi^2(p)$; E(X) = p, Var(X) = 2p

(iid=identically Independent distributed)

4. t-분포 : r.v. Z ~ N(0,1), r.v. V~χ(ν)

- $X = \frac{Z}{\sqrt{V/\nu}}$ ~ t-분포; $E(x) = 0, Var(X) = \frac{\nu}{\nu-2}$ $(\nu > 2)$; if ν→∞,(t-분포) →N(0,1)
  Var >1 → S.N.과 비교 시 더 큰 불확실성

5. F-분포 : let A,B → independent r.v., $A \sim \chi^2(r_1), B \sim \chi^2(r_2) \rightarrow F = \frac{A/r_1}{B/r_2} \sim F(r_1, r_2)$

5. 중심극한정리(CLT; Central limit theorem)

- 분포수렴(convergence in distribution) : $X_1, \cdots, X_n, X_{n+1} \rightarrow r.v.\ X$의 분포로 수렴한다
  → $X_n \xrightarrow{d} X$: $\lim_{n\to\infty} P(X_n \le x) = P(X \le x)$ ∀(x | Fx가 연속). 즉, CDF 수렴→ 분포 수렴
- $CLT: r.v. (X_1, \cdots, X_n, \cdots), E(X) = \mu, Var(X) = \sigma^2$인 iid r.v. $\overline{X_n} = \frac{1}{n}\sum X_i$, $\frac{\sqrt{n}(\overline{X_n}-\mu)}{\sigma} \xrightarrow{d} N(0,1)$

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